Modular and Auditable Clinical Extraction from Epilepsy Letters

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***Abstract*—We evaluate a modular clinical extraction architecture on two complementary epilepsy-letter tasks: deep seizure-frequency extraction and broad clinical phenotyping. The first task tests temporal reasoning and benchmark-facing normalization on the Gan 2026 seizure-frequency corpus; the second tests multi-entity recovery on ExECTv2 using de-duplicated clinical-headline facts as the primary surface. Across both tasks, the system separates model-mediated clinical interpretation, deterministic normalization and projection, evidence-grounding checks, reliability audits, and component-impact analyses. This draft reports aggregate validation and frozen holdout evidence under fixed inspection boundaries, and it deliberately separates reliability scorecards from component-causal claims.**

***Index Terms*—clinical information extraction, epilepsy, large language models, hybrid systems, reliability, component ablation**

I. Introduction

Clinical extraction systems must often trade breadth, depth, auditability, and portability. Epilepsy letters make the tension concrete: seizure-frequency statements require temporal anchoring, seizure-free duration handling, current versus historical selection, and benchmark-compatible normalization, while broader phenotyping requires the same infrastructure to recover diagnosis, medication, investigation, and seizure-frequency facts without collapsing into a single task-specific rule stack. Prior epilepsy NLP systems have shown both the value of broad clinic-letter extraction and the persistent difficulty of seizure-frequency concepts [1], [2], while recent seizure-frequency work emphasizes pretrained models, LLM fine-tuning, and synthetic but task-faithful supervision [3]–[6].

This paper studies a modular extraction design in which deterministic rules and large language model (LLM) components are explicit, testable parts of the scientific object. We evaluate the architecture on a deep single-concept task (Gan 2026 seizure-frequency extraction) and a broad multi-entity task (ExECTv2 clinical-fact recovery). The central claim is not that any one model or rule set is universally optimal, but that explicit component boundaries make it possible to report performance, reliability, and component impact without blending incompatible evidence surfaces.

II. Related Work

Prior epilepsy natural language processing work includes broad phenotyping systems and narrow seizure-frequency extraction systems. Recent LLM-based clinical extraction work has improved coverage and flexibility, but can obscure the role of deterministic preprocessing, schema repair, normalization, and post-processing [4], [5]. The present draft positions deterministic rules as controlled variables rather than incidental implementation detail, and treats reliability audits and component-impact replays as separate manuscript objects. This emphasis is aligned with evidence-grounded seizure-frequency supervision in recent synthetic-letter work, but extends the reporting discipline to component boundaries and aggregate replay evidence [6].

III. Methods

A. Tasks and Evaluation Surfaces

Gan 2026 is used as the deep seizure-frequency task [6]. Development follows a locked split protocol: validation rows are used for development, while `test450` is reserved for frozen final aggregate evaluation. Test-set row-level failure inspection is not used for development or tuning.

ExECTv2 is used as the broad phenotyping task, building on the ExECT lineage of epilepsy clinic-letter extraction and the later synthetic-letter annotation resource [1], [2]. The headline surface is de-duplicated `clinical_headline` recovery across Diagnosis, SeizureFrequency, Prescription, and Investigations. Strict benchmark and CUI-oriented outputs are retained as diagnostic compatibility surfaces, not as the primary success criterion.

B. Architecture Families

We compare rules-only, LLM-only, and hybrid families. In Gan 2026, the promoted hybrid structured-event architecture uses an LLM to extract source-near seizure events and deterministic code for normalization, projection, rendering, and scoring. In ExECTv2, a manifest-driven clinical-finding assembly combines candidate producers, entity-specific reconciliation lenses, evidence validation, and scoring views.

C. Evidence Boundaries

We distinguish four evidence types throughout the paper. Architecture comparison describes aggregate task performance under a fixed scorer. Frozen holdout evidence reports locked evaluation outputs without authorizing post-hoc tuning. Reliability scorecards test whether a fixed system is grounded, calibrated, robust, consistent, and operationally replayable.

Component-impact analysis is reserved for ablations or same-input replays that measure a named component's score contribution.

IV. Results

We evaluate the shared modular architecture on two complementary epilepsy-letter tasks: deep seizure-frequency extraction (Section IV-A) and broad nine-entity phenotyping (Section IV-B). For each task we report architecture comparison, frozen holdout or aggregate validation evidence under declared inspection boundaries, and, where predeclared, separate reliability and component-impact readouts.

A. Gan 2026 Seizure-Frequency Extraction

1) Three-Way Architecture Comparison: On validation750, `hybrid_structured_events` combined the best LLM-using coverage and accuracy balance among the canonical families: the LLM extracts source-near structured events with exact evidence spans, and deterministic code owns normalization, projection, and scoring. The deterministic canonical pipeline led validation after de-overfitting but showed the largest validation-to-holdout drop on the frozen `test450` audit, supporting the portability warning that high validation score from rules alone is incomplete generalization evidence. The validation750 table is retained as supporting development evidence in Appendix Table V.

The promoted close-off candidate is the single GPT structured-event pass on `gpt-4.1-mini`, which reached 364/450 Purist (0.809) on locked `test450` with the smallest validation-to-test drop among LLM-using architectures. The full V12 fresh-evidence hybrid reached 379/450 Purist (0.842) but is retained only as a high-complexity ceiling comparator, not the operational headline system.

2) Frozen Holdout Aggregate Evidence: Late-cycle consensus/fresh selector evidence is reported only as frozen aggregate audits completed under predeclared Gate 4 protocols. The constrained-source audit failed promotion bars; the exact-source audit passed promotion bars. Neither audit authorizes post-test tuning or row-level failure inspection. The audit rows are appendix evidence because they are late-cycle selector diagnostics rather than the promoted operational architecture (Appendix Table VI).

Gan holdout-facing claims are limited to aggregate `test450` numbers under frozen protocols. Validation750 rows are development evidence. The structured-event hybrid leads the frozen three-way comparison, while the exact-source consensus/fresh selector records a bounded +16 Purist holdout gain at 0.6000 changed-label precision. These results do not authorize further gate, prompt, or selector tuning from locked-test aggregates.

3) Bridge to ExECTv2: Seizure-frequency normalization, temporal anchoring, and seizure-free handling developed for Gan 2026 are the direct substrate for ExECTv2 SeizureFrequency recovery. The cross-task read is not a benchmark-win claim: it tests whether the modular investment transfers to a new annotation schema and scoring surface. ExECTv2 SeizureFrequency remains the weakest family on `clinical_headline` despite positive full-200 aggregate recovery, consistent with the deep-reasoning difficulty established by Gan 2026.

B. ExECTv2 Clinical-Fact Recovery

1) Architecture and Clinical-Headline Performance: We evaluate the selected ExECTv2 clinical-finding assembly primarily by de-duplicated clinical-headline recovery rather than strict full-schema annotation reproduction. On the aggregate full-200 current-code audit, the verifier-backed GPT-4.1-mini v08-shaped architecture scored 0.8502 overall clinical-headline F1. The accepted lean candidate, the 2-call no-SF-adjudicator GPT-4.1-mini system, reduced the full-200 call profile to 400 calls while preserving the governing cost-performance gates: 0.8356 overall and 0.7525 SeizureFrequency F1.

Strict benchmark and CUI-oriented outputs measure compatibility with the original annotation surface. They are reported only as secondary diagnostics and are not described as the primary ExECTv2 success criterion.

2) Same-Core Model Swap: Using the frozen `exectv2_2call_no_sf_adjudicator_model_swap` core, DeepSeek chat, GPT-4.1-mini, and Qwen 3.6 35B were compared under the same component graph and `clinical_headline` scorer. On dev140, DeepSeek produced the strongest aggregate score, GPT-4.1-mini remained operationally clean, and the unrepaired Qwen row remained diagnostic because output-contract failures blocked operational promotion. Qwen repair v02 later passed the predeclared dev140 and full-200 aggregate gates with zero call/parse failures in both assemblies. Detailed model-swap rows move to the appendix because they support model-family robustness rather than the central architecture comparison (Appendix Tables VII and VIII).

The same-core model-swap study suggests that the ExECTv2 component graph is not GPT-4.1-mini-specific. DeepSeek reached 0.8596 and GPT-4.1-mini reached 0.8396 clinical-headline F1 on dev140 under the frozen core, with full-200 aggregate rows of 0.8566 and 0.8356 respectively. Qwen repair v02 provides additional same-core model-family evidence after output-contract repair, but remains below the operational GPT-4.1-mini and DeepSeek rows. The unrepaired Qwen dev140 row is retained as a diagnostic comparator only.

3) Reliability Scorecard: The reliability scorecard follows the main performance tables and tests whether the selected ExECTv2 system behaves faithfully under fixed scoring and inspection boundaries. Current evidence supports aggregate full-200 reliability claims with no holdout or deployment-probability claim.

The reliability scorecard tests whether the selected ExECTv2 system behaves faithfully under fixed scoring and inspection boundaries. Calibration and robustness have predeclared aggregate full-200 evidence; review routing has a useful high-recall diagnostic but no validated low-burden policy; and consistency is supported by saved live-repeat panels for the selected lean candidate. These results support trust in the fixed

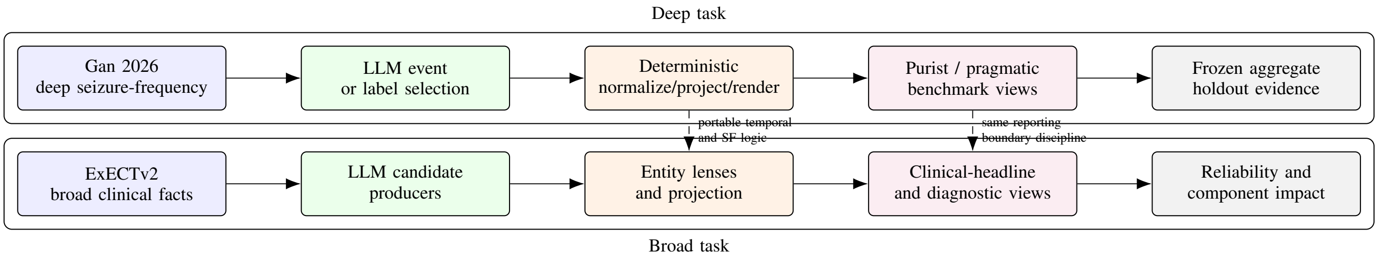

Fig. 1. Two-task modular evidence architecture. Gan 2026 stresses temporal seizure-frequency extraction under frozen aggregate holdout rules; ExECTv2 reuses the same separation of model-mediated candidates, deterministic clinical logic, scoring views, and reliability/component-impact evidence for broad clinical-fact recovery.

TABLE I
GAN 2026 FROZEN `TEST450` AGGREGATE AUDIT (`GPT-4.1-MINI`)

Architecture	Rendered	Purist	Pragmatic	Claim use
`deterministic_canonical_pipeline`	450/450	329/450 (0.731)	341/450 (0.758)	Frozen aggregate; validation-to-test gap warning.
`hybrid`	334/450	269/334 (0.805)	281/334 (0.841)	Frozen aggregate; coverage-limited.
`hybrid_structured_events`	448/450	364/448 (0.812)	381/448 (0.850)	Strongest frozen hybrid aggregate on rendered rows.
`llm_only_canonical_pipeline`	450/450	326/450 (0.724)	346/450 (0.769)	Frozen comparator row.

TABLE II
EXECTV2 ARCHITECTURE COMPARISON ON FULL-200 AGGREGATE `CLINICAL_HEADLINE`

System/view	Overall	Diagnosis	SF	Prescription	Investigations	Claim use
Current-code v08-shaped GPT-4.1-mini	0.8502	0.8321	0.7850	0.8926	0.9213	Aggregate architecture evidence.
No-verifier ablation	0.8431	0.8410	0.7850	0.8926	0.8563	Component-role comparison.
Accepted lean 2-call no-SF-adjudicator	0.8356	0.8397	0.7525	0.8926	0.8563	Cost-performance frontier.

TABLE III
EXECTV2 RELIABILITY SCORECARD

Dimension	Current evidence	Manuscript claim
Evidence grounding	Same-core rows report minimum exact evidence rate 1.0000; robustness audit has schema/evidence validity 1.0000.	Outputs are evidence-grounded on audited surfaces.
Calibration	Frozen grouped scoring rule validates on full-200 with ECE 0.0432, Brier 0.2245, and base-rate Brier 0.2387.	Aggregate calibration evidence.
Review routing	Lower-burden candidate failed validation because burden rose to 0.9661 despite 0.9037 catch.	No promoted low-burden triage policy.
Robustness	Current-code v08 hard-slice F1 is 0.8336 across 414 eligible family cells versus 0.8503 overall.	Aggregate hard-slice validation evidence.
Consistency	Lean candidate has hard50 temp-0 exact agreement 0.9217 and dev140 varying-temperature exact agreement 0.8857.	Live-repeat consistency evidence.
Component role	Investigations verifier plus deterministic suppression remains strongest at 0.9213; deterministic replacement is not ready.	Rules are verification/suppression aids.

architecture but are not holdout performance claims and are not component-causal ablation evidence.

4) Component Impact: Component impact is reported separately from the reliability scorecard. Claims are limited to aggregate replay deltas under a fixed scorer, split, and inspection boundary. Dev140 one-component-off replay and full-200 aggregate-only replay support limited component-impact language for dictionary normalization, residual semantic recovery, and headline projection on `clinical_headline`.

The no-verifier full-200 ablation and Investigations rule ablation remain separate component-role comparisons. They should not be blended with the layer-ladder component-off readouts above. Component-impact claims are reserved for ablations and same-input replay deltas under a declared scorer and inspection boundary. Dictionary, residual-semantic, and headline-projection layers show positive aggregate deltas on dev140 and full-200 replay, but these results do not prove any component is globally required.

V. DISCUSSION

The two-task readout supports the value of modular, auditable extraction over a single monolithic score. Gan 2026 demonstrates that seizure-frequency extraction is a high-pressure temporal reasoning task: deterministic rules can reach high validation scores but show a pronounced holdout gap, while hybrid structured events provide the strongest frozen LLM-using aggregate evidence. ExECTv2 tests whether the same design discipline scales to broader clinical-fact recovery,

TABLE IV
COMPONENT-OFF AGGREGATE DELTAS ON `clinical_headline`

Component	Type	Split	Overall delta	Main family signal	Claim boundary
`standard_dictionary`	dictionary	dev140	+0.0389 to +0.1120	Diagnosis/SF gains	Conditional dictionary recovery.
`standard_dictionary`	dictionary	full200	+0.0186 to +0.0290	Diagnosis up to +0.0802	Same-core full-200 impact only.
`residual_semantic_lens`	semantic lens	dev140	+0.0175 to +0.1041	Investigations gains	Prediction-bearing semantic contribution.
`residual_semantic_lens`	semantic lens	full200	+0.0098 to +0.0117	Diagnosis up to +0.0310	Aggregate replay only.
`headline_projection`	deterministic projection	dev140	+0.0283 to +0.0446	SF up to +0.2031	Projection/format contribution.
`headline_projection`	deterministic projection	full200	+0.0302 to +0.0350	SF up to +0.1417	Format layer only.

where `clinical_headline` provides the headline surface and strict benchmark/CUI outputs remain diagnostic.

The most important reporting constraint is the separation of reliability and component impact. Reliability scorecards say whether a fixed architecture is grounded, calibrated, robust, consistent, and replayable under the declared surface. They do not prove that an individual component caused the observed performance. Component-impact claims require ablations or same-input replays, and in this draft remain bounded to the declared scorer, split, and inspection policy.

A. Limitations

Gan 2026 locked-test evidence is aggregate-only and does not authorize row-level test debugging or post-test tuning. ExECTv2 full-200 evidence is also aggregate-only under the current reliability and component-impact protocols. The ExECTv2 `clinical_headline` surface is intentionally not a strict benchmark win claim. Calibration and review-routing results are not deployment readiness claims, and Qwen repair v02 is model-family evidence rather than an operational promotion over GPT-4.1-mini or DeepSeek.

VI. CONCLUSION

Across deep seizure-frequency extraction and broad epilepsy-letter phenotyping, the modular architecture enables performance reporting, reliability assessment, and component-impact analysis without conflating their evidence boundaries. The current results support aggregate, claim-bounded conclusions: hybrid structured events are the strongest frozen Gan 2026 close-off candidate, and ExECTv2 achieves strong deduplicated clinical-headline recovery with separate reliability and component-impact evidence.

APPENDIX A
SUPPORTING RESULT TABLES

The main text keeps four result tables: the Gan 2026 frozen aggregate audit, the ExECTv2 architecture comparison, the ExECTv2 reliability scorecard, and the ExECTv2 component-impact table. The remaining four tables are supporting evidence: validation development context, late-cycle selector diagnostics, and same-core model-family robustness.

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TABLE V
GAN 2026 THREE-WAY COMPARISON ON VALIDATION750 (GPT-4.1-MINI)

Architecture	Rendered	Purist	Pragmatic	Reading
deterministic_canonical_pipeline	741/750	673/741 (0.908)	681/741 (0.919)	High validation score; large holdout drop.
hybrid	597/750	526/597 (0.881)	545/597 (0.913)	Strong rendered accuracy; too many null/routed rows.
hybrid_structured_events	748/750	661/748 (0.884)	679/748 (0.908)	Best LLM-using validation coverage/accuracy balance.
llm_only_canonical_pipeline	750/750	582/750 (0.776)	614/750 (0.819)	Comparator; below hybrid structured events.

TABLE VI
GAN 2026 CONSENSUS/FRESH V0.9 FROZEN GATE 4 AGGREGATE AUDITS (TEST450)

Audit	Source symmetry	Selected Purist	Net gain	Changed precision	Gate outcome
Constrained	constrained	348/450 (0.7733)	+19	0.5909	Failed; final-evaluation evidence only.
Exact-source	exact	359/450 (0.7978)	+16	0.6000	Passed; frozen exact selector holdout.

TABLE VII
SAME-CORE MODEL SWAP ON DEV140 CLINICAL_HEADLINE

Candidate/model	Overall	Diagnosis	SF	Prescription	Investigations	Operational caveat
DeepSeek chat	0.8596	0.8845	0.7658	0.8895	0.8966	One parse/schema failure.
GPT-4.1-mini	0.8396	0.8573	0.7645	0.8895	0.8347	Operationally clean.
Qwen 3.6 35B repair v02	0.8319	0.8473	0.7182	0.8895	0.8755	Passes repair gates.
Qwen 3.6 35B unrepaired	0.8018	0.8027	0.6919	0.8895	0.8354	Diagnostic only.

TABLE VIII
SAME-CORE MODEL SWAP ON FULL-200 AGGREGATE CLINICAL_HEADLINE

Candidate/model	Overall	Diagnosis	SF	Prescription	Investigations	Call/parse failures
DeepSeek chat	0.8566	0.8708	0.7602	0.8926	0.9091	0 / 1
GPT-4.1-mini	0.8356	0.8397	0.7525	0.8926	0.8563	0 / 0
Qwen 3.6 35B repair v02	0.8197	0.8307	0.7020	0.8926	0.8503	0 / 0